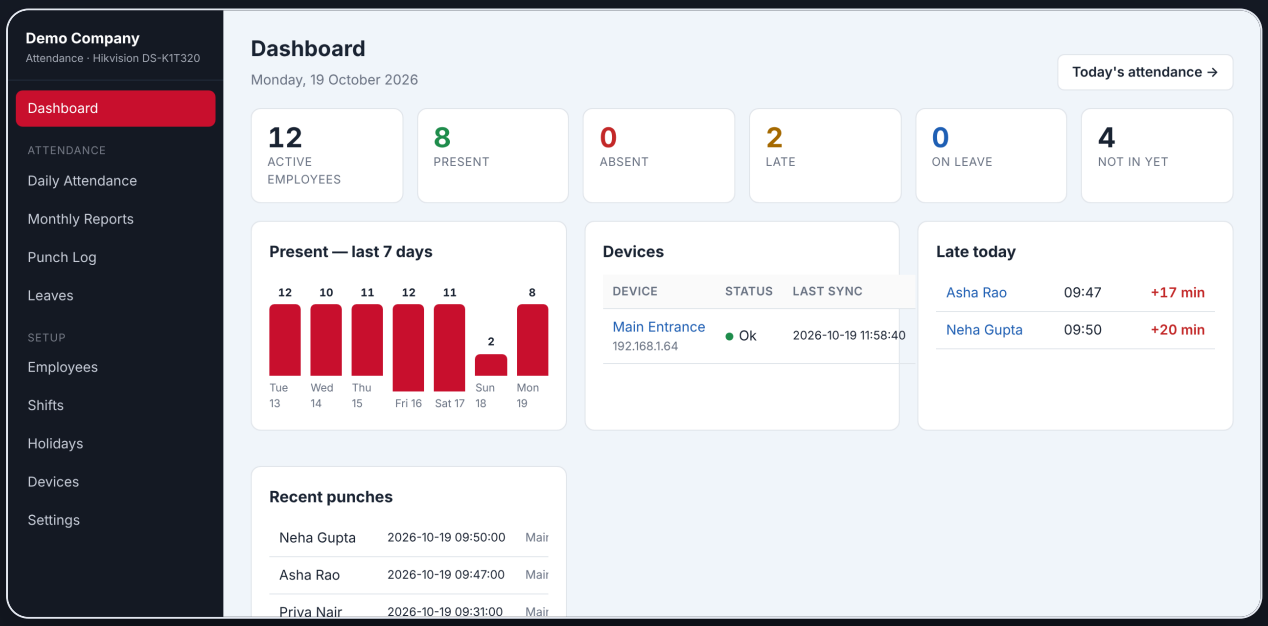
COMPLETE GUIDE · VERSION 1.0

Attendance Management System

For the Hikvision DS-K1T320EFWX face-recognition terminal. What the app does, how to use it, what's done and what's next.



Web app · works on PC and mobile

October 2026

What's inside

This guide is written for owners, HR and administrators, not programmers. Each part stands on its own, so skip to what you need.

1	**The big picture**	What the system is, how the pieces connect
2	**Feature tour**	Every screen, with annotated screenshots
3	**How attendance is calculated**	Present, late, half day, overtime: the rules explained
4	**What's done**	Everything built and tested so far
5	**What needs to be done**	Your go-live checklist, and optional future features
6	**Keeping it running**	Updates without losing data, backups, daily routine
7	**Help & glossary**	Common problems and plain-English terms

How to read the status labels

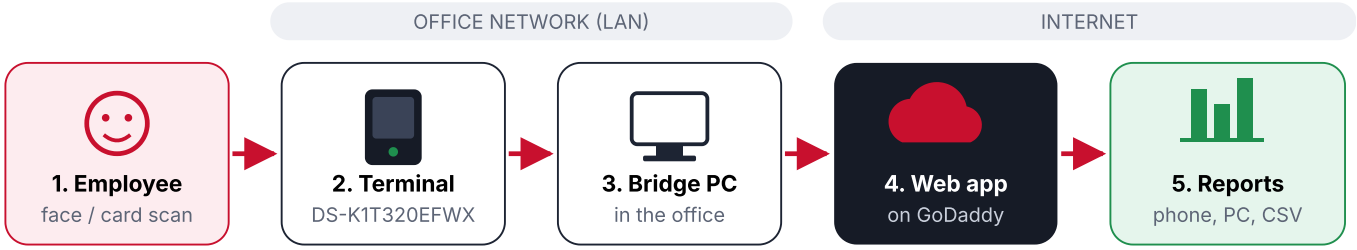
DONE	Built, working and covered by automated tests.
YOUR ACTION	Something only you can do, such as buying hosting, setting up the office PC or entering passwords.
VERIFY ON DEVICE	Built and tested against a simulated terminal. Confirm it on your real DS-K1T320EFWX during the pilot.
FUTURE (OPTIONAL)	An idea for later. Not built yet and not needed to go live.

Where the technical details live. Step-by-step hosting instructions are in `docs/DEPLOY_GODADDY.md` in the code repository. This PDF summarises them and links them to the bigger picture.

1 The big picture

Employees show their face (or card) at the Hikvision terminal. Every scan is collected automatically. The app turns the scans into attendance, late marks, overtime and monthly reports that you open in any web browser.

From a face scan to a monthly report



Punches reach the web app within about 5 minutes (the interval is set in Settings).

Why is there a "bridge PC"?

The terminal has a private office address (for example `192.168.1.64`). A website on the internet cannot reach a private address. The small **bridge program** runs on any always-on office PC and passes data both ways:

- **Up:** new punches from the terminal go to the website.
- **Down:** actions you click on the website (send employee, enrol face, open door, set clock) go to the terminal.

It only makes outgoing connections, so you need no router changes, port forwarding or static IP.

Who uses it

ROLE	CAN DO
Admin	Everything: employees, shifts, devices, corrections, settings, users.
Viewer	Read-only: dashboards, attendance and reports. Suits managers and accounts staff.
Employees	Only scan at the terminal. They have no login FUTURE

In one minute

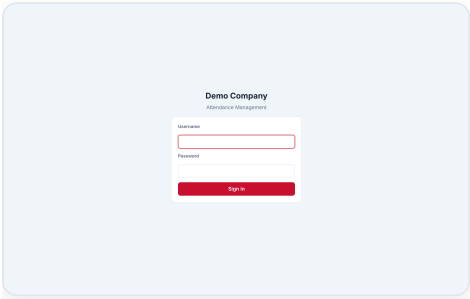
12	7	6
SCREENS	ATTENDANCE STATUSES	CSV EXPORTS + PRINT

- **Automatic:** punches arrive by themselves. Nobody types attendance.
- **Always correct:** attendance is recalculated from the raw scans every time you open a report. Fix a shift, add a leave or a missed punch, and every report updates straight away.
- **Safe:** passwords are stored encrypted, nightly backups are kept, and updates never reset your data.
- **Anywhere:** works in any browser on a PC, tablet or phone.

2 Feature tour

A screen-by-screen walk through the app. The numbered red pins on the screenshots match the notes underneath.

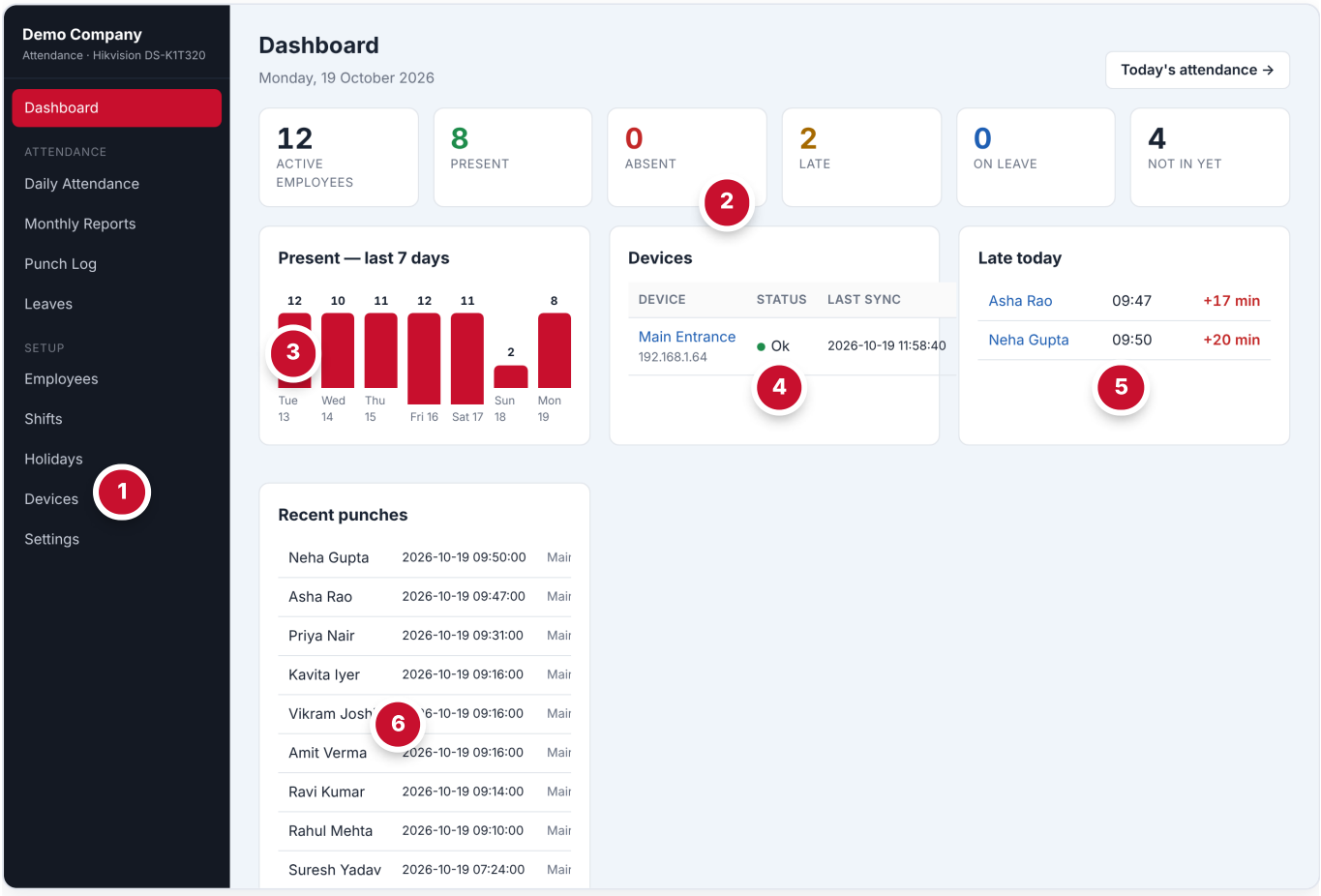
2.1 Signing in DONE



Everyone signs in with a username and password created by an admin in **Settings → Users**.

- The first login is `admin` / `admin`. The app reminds you to change it, so change it straight away.
- Passwords are stored encrypted, never as plain text.
- Forms are protected against cross-site attacks (CSRF), and sessions use HTTPS-only cookies once the site is on HTTPS.
- A lost admin password can be reset from the server (see Part 7).

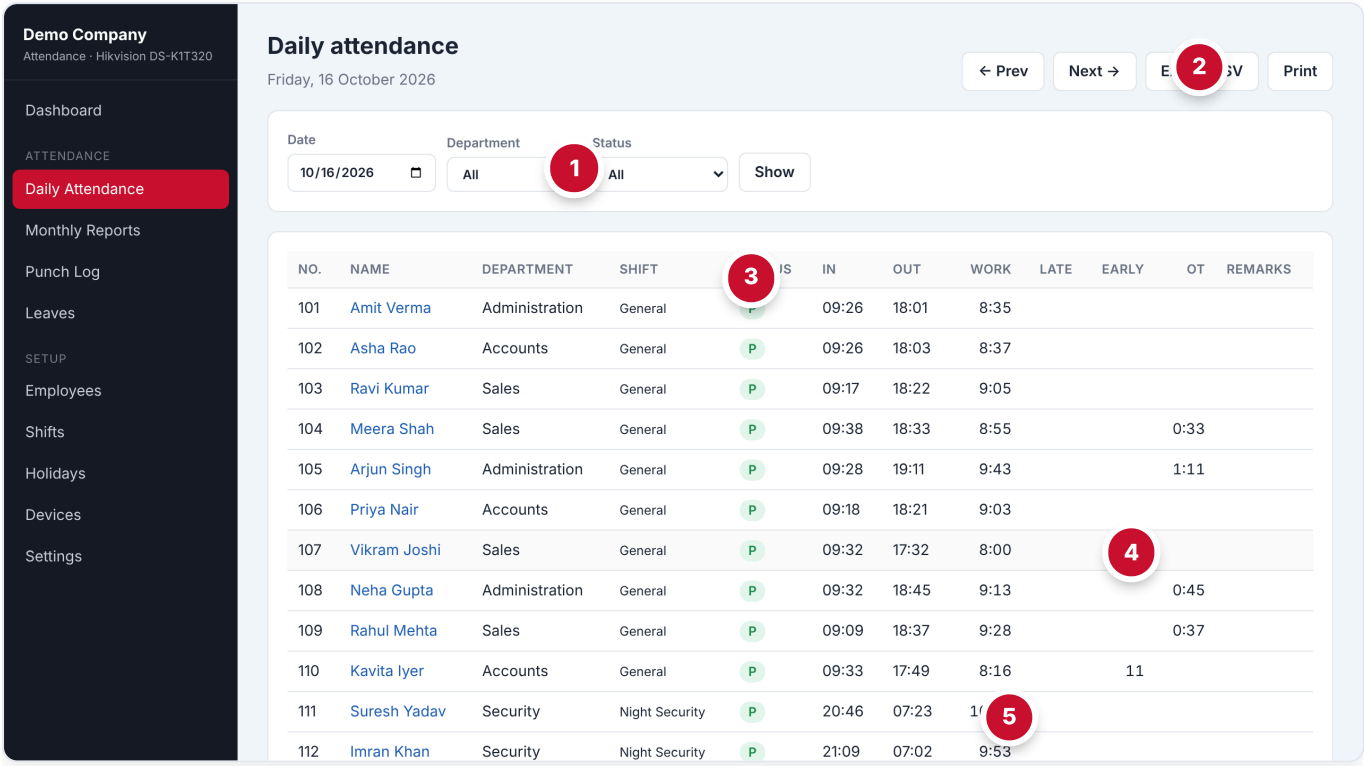
2.2 Dashboard: today at a glance DONE



1. **Menu.** Attendance screens at the top, setup screens below. On phones it folds into a "Menu" button.
2. **Today's counts:** present, absent, late, on leave, and "not in yet" (their shift hasn't ended).
3. **7-day trend** of how many people came in each day. Sundays and holidays show as dips.
4. **Device health:** a green dot means the terminal is syncing. Red means look at the Devices page.
5. **Late today**, with the arrival time and minutes late.
6. **Recent punches** as they arrive from the terminal.

2.3 Daily attendance DONE

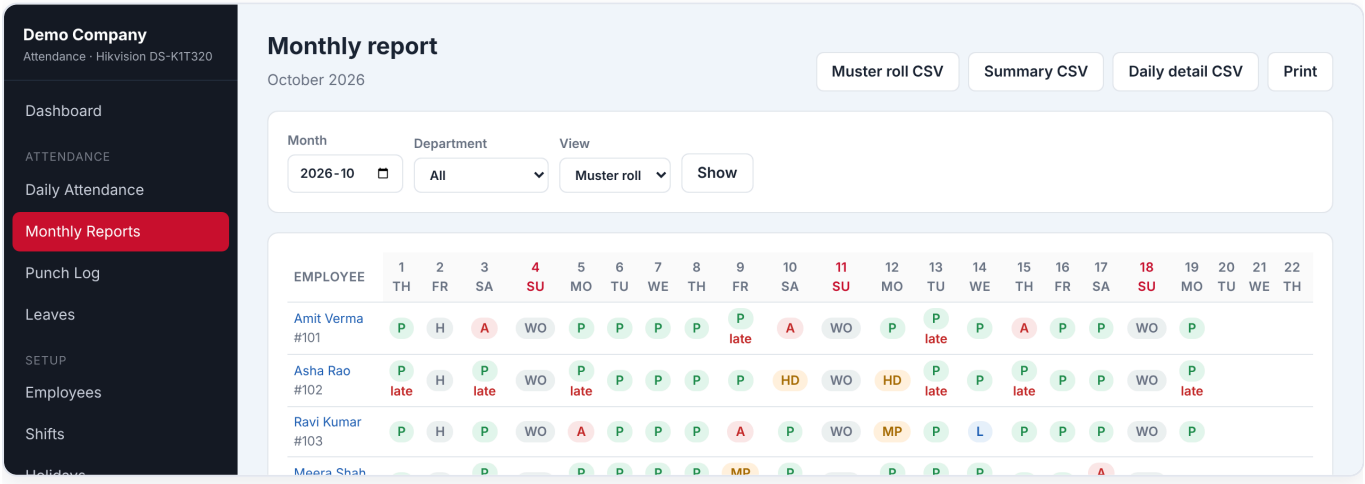
One row per employee for any date: in-time, out-time, hours worked, late and early minutes, overtime and status. Use **Prev / Next** to move through the days.



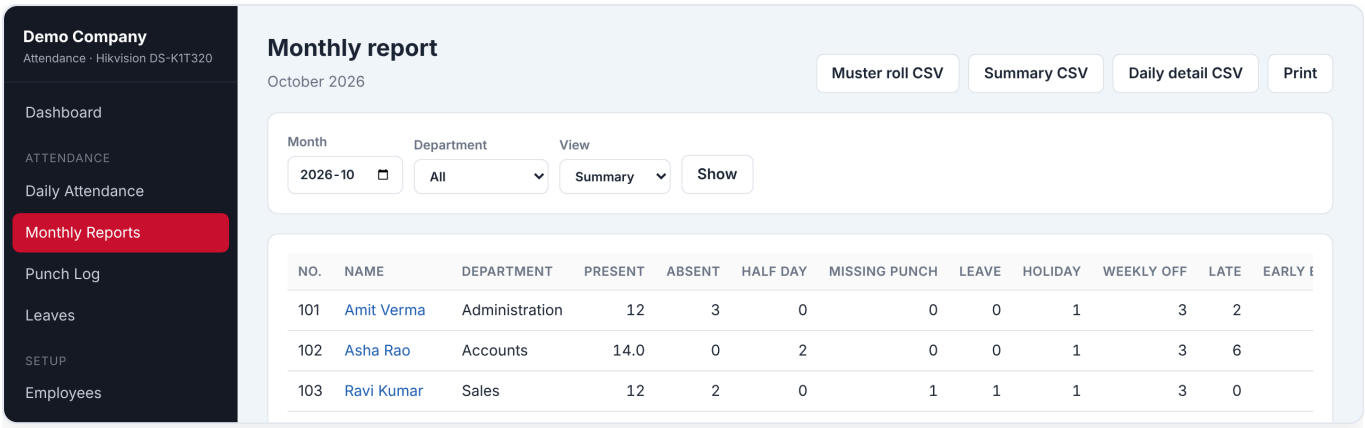
1. **Filters:** pick a date, a department, or a status (for example "Late" or "Absent" only).
2. **Export CSV** opens in Excel. **Print** gives a clean paper version.
3. **Status badge**: P, A, HD, MP, L, H, WO (explained in Part 3).
4. **Late / Early** in minutes, and **OT** (overtime) as hours:minutes.
5. **Night shifts** work correctly: in at 20:46, out at 07:23 the next morning counts as one day.

2.4 Monthly reports DONE

Two views of a whole month, each with a CSV download for payroll:



Muster roll: every employee × every day, colour-coded. Hover over a badge on screen to see that day's in/out times. A red "late" marks a late arrival. Sundays are in red in the header.

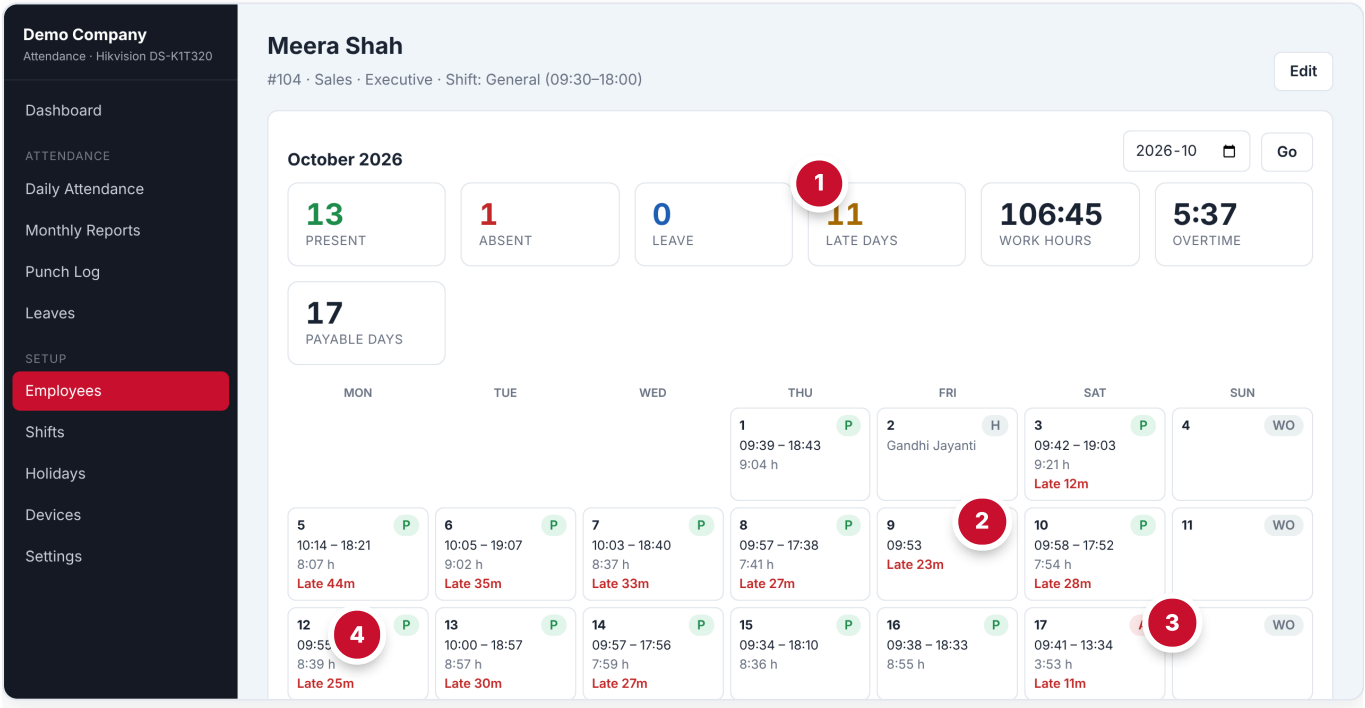


Summary view: per employee, days present, absent, half days, missing punches, leave, holidays, weekly offs, late days, early exits, work hours, overtime hours and **payable days**.

DOWNLOAD	WHAT YOU GET	TYPICAL USE
Muster roll CSV	Status code for each day + monthly totals	Records, inspections, labour compliance
Summary CSV	One row per employee with all totals	Salary calculation
Daily detail CSV	One row per employee per day with times	Detailed checks, disputes

2.5 Employee profile & calendar DONE

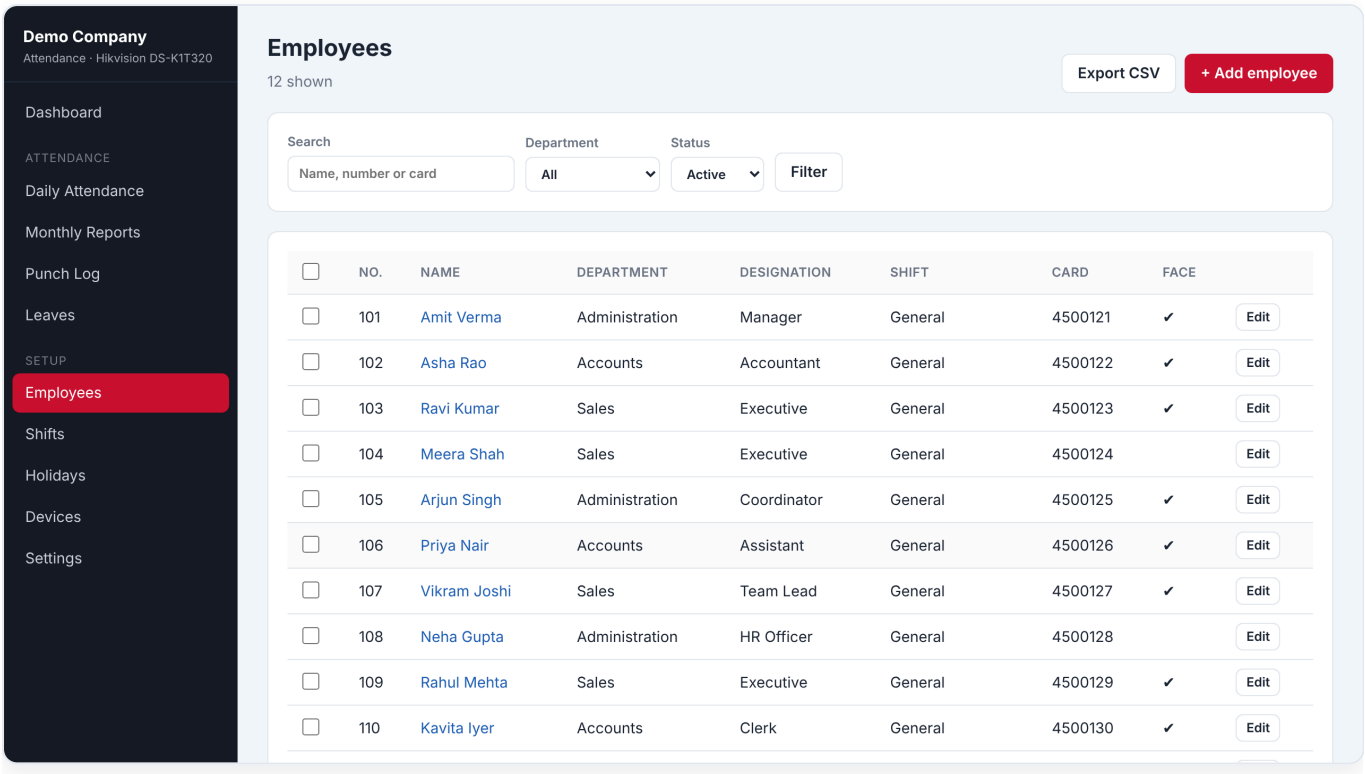
Click any name to see that person's month as a calendar, with totals on top.



1. **Month totals:** present, absent, leave, late days, hours, overtime, payable days.
2. **MP = missing punch**: only an in-time was recorded. Fix it by adding a manual punch.
3. **A after a short day:** worked under the half-day minimum (3:53 h here).
4. Each day shows in–out, hours worked and late minutes.

Below the calendar (not shown): employee details, plus **Device enrolment** to send the user and card to the terminal, upload a face photo, add a manual punch, or remove the employee.

2.6 Employees DONE



Ways to add people

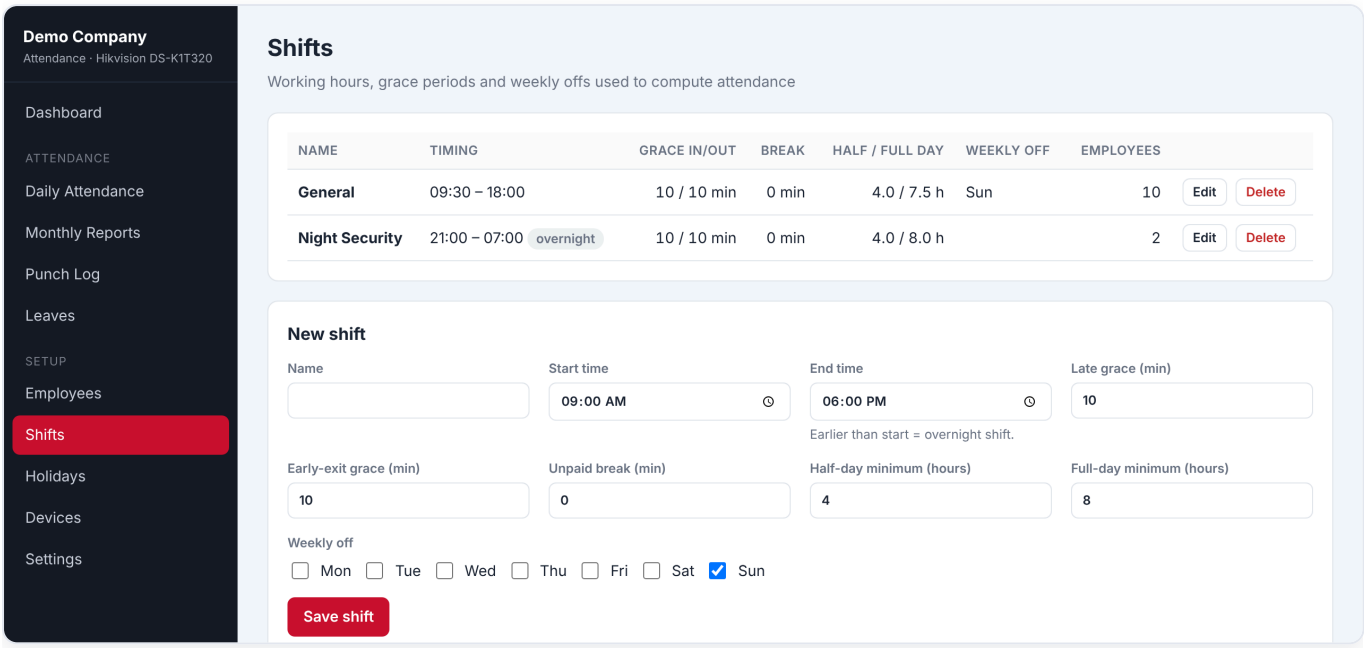
- **Import users from device:** pulls everyone already enrolled on the terminal.
- **Add employee:** one at a time, and optionally sends them to the terminal at the same time.
- **Import CSV:** many at once from Excel (columns `employee_no, name, department, …`).

Handy tools

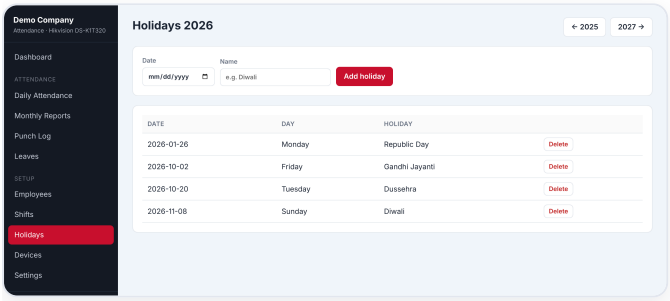
- Search by name, number or card. Filter by department or active/inactive.
- **Bulk change:** tick employees, then set a shift or department for all of them.
- **Deactivate** instead of deleting, to keep a leaver's history.
- The **Face** column shows who has a face enrolled.

> **Important:** the **Employee No.** in the app must be the same as the **Employee ID** on the terminal. That is how a scan is matched to a person.

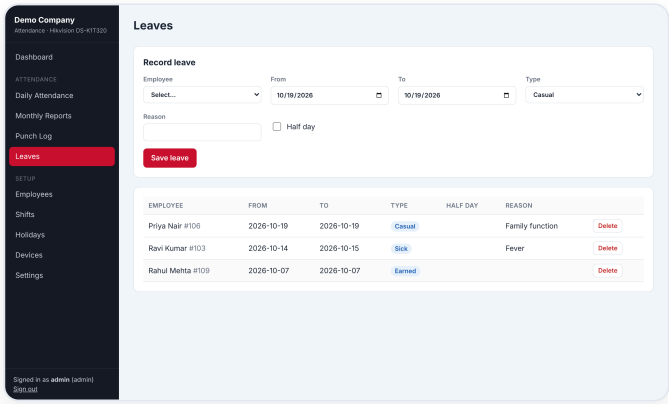
2.7 Shifts, holidays and leave DONE



Shifts set the working hours, grace period for late and early, unpaid break, half-day and full-day hours, and weekly offs. A shift whose end time is earlier than its start time is treated as a **night shift**.

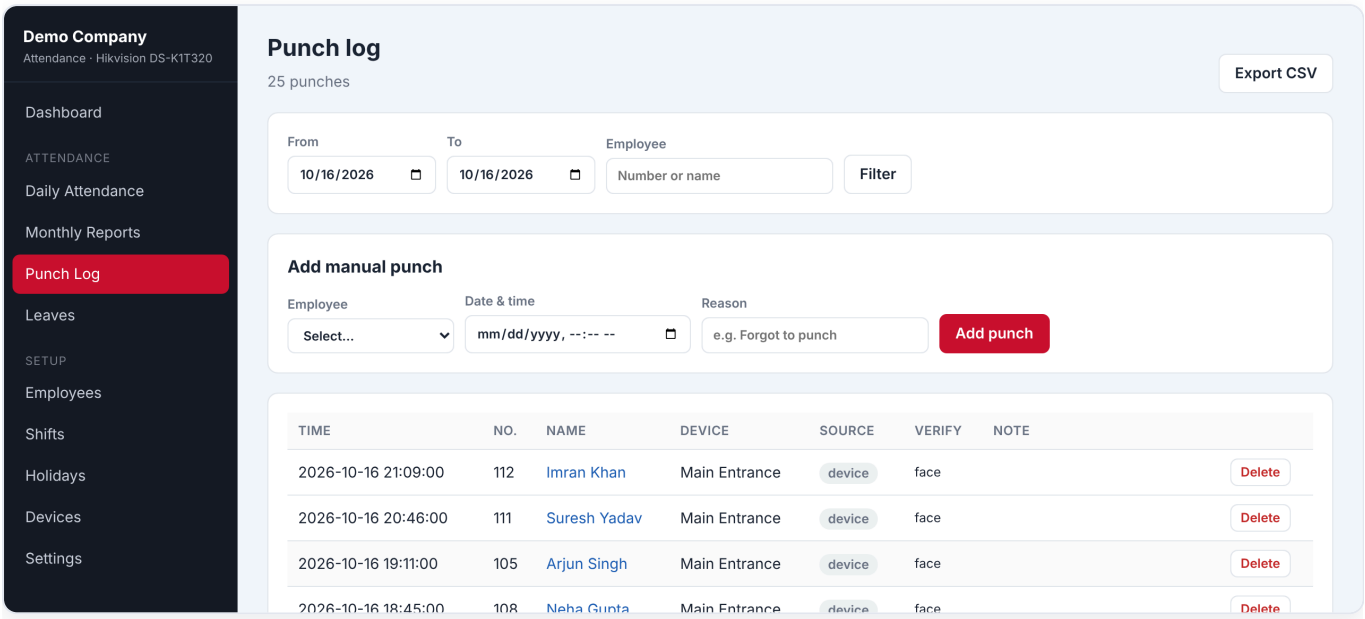


Holidays: add the year's public holidays once. Those days show as H instead of absent. Anyone who works on one gets the hours as overtime.



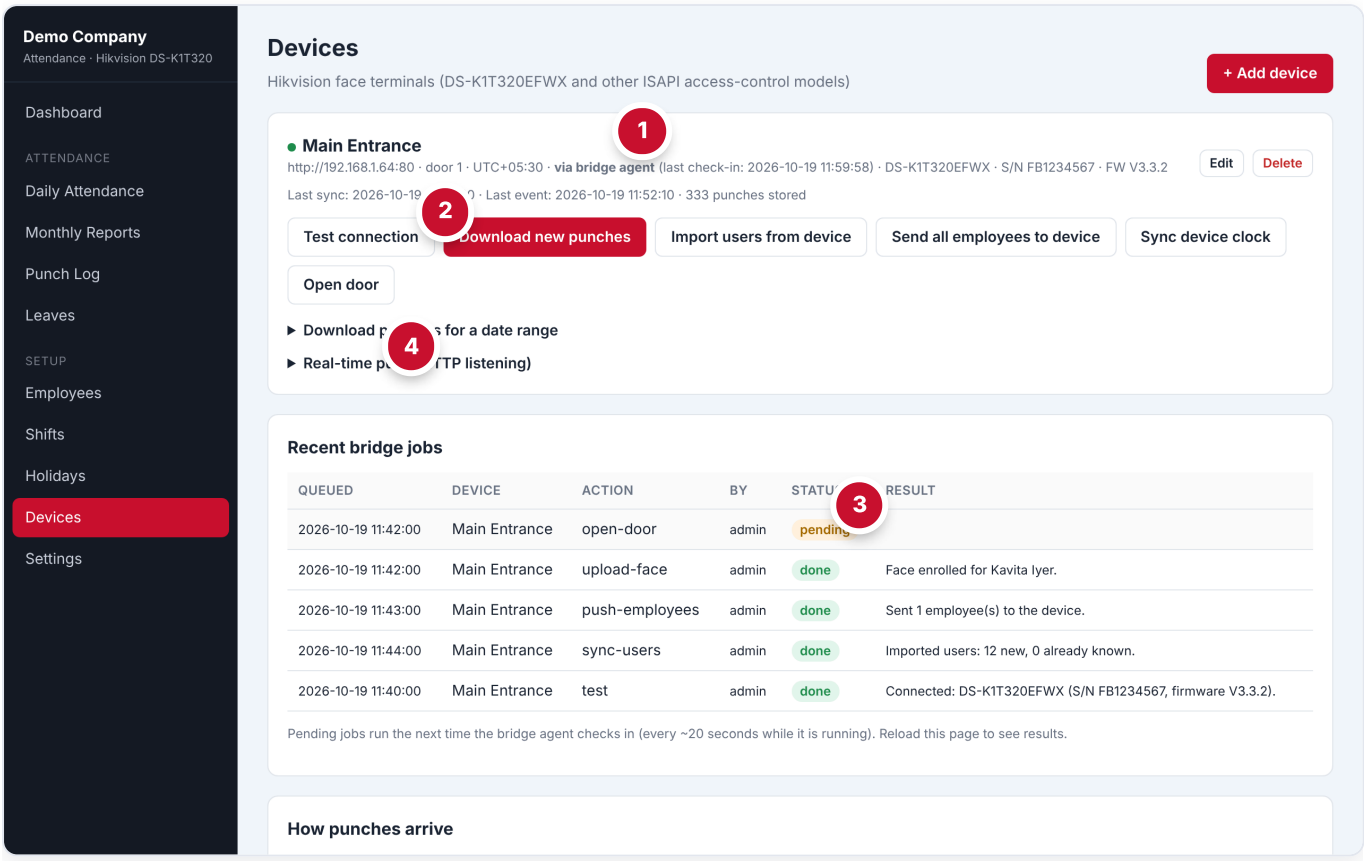
Leaves: record full or half-day leave (Casual, Sick, Earned, Unpaid, Comp Off…). Leave days show as L and count as payable.

2.8 Punch log & corrections DONE



Every raw scan, with the device and how it was verified (face or card). An admin can **add a manual punch**, for example when someone forgot to punch out. Manual punches are labelled with the reason and the admin's name, so there is an audit trail. Wrong punches can be deleted.

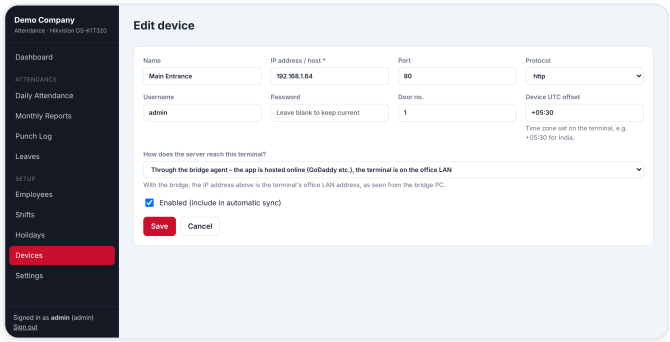
2.9 Devices & the bridge DONE VERIFY ON DEVICE



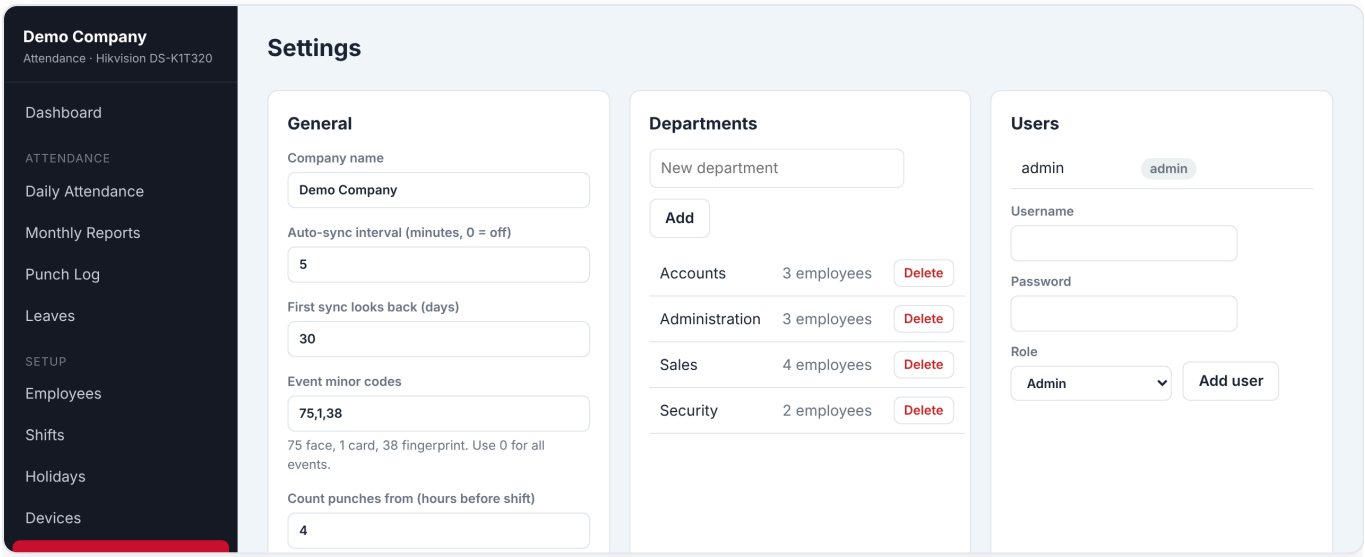
1. **Connection:** "via bridge agent", with the bridge's last check-in time, so you can see it's alive.
2. **One-click actions:** test, download punches, import or send users, set the clock, open the door.
3. **Job list:** each click becomes a job. The bridge picks it up within about 20 seconds and reports `done` or failed, with the reason.
4. **Download a date range** to recover older punches, for example after a long power cut.

Adding the terminal

1. Devices → **Add device**.
2. Enter the terminal's office IP (e.g. 192.168.1.64), its admin username and password, and the time zone (+05:30).
3. Set **Connection** to *Through the bridge agent*.
4. Click **Test connection**, wait about 20 s, then reload.

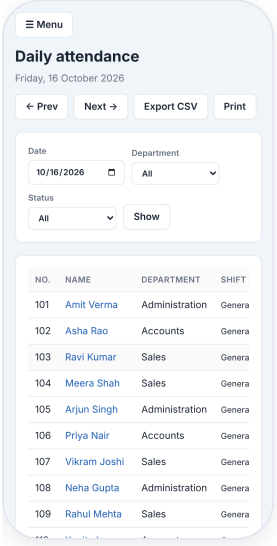


2.10 Settings, users & backup DONE



SETTING	WHAT IT DOES	DEFAULT
Company name	Shown in the menu, page titles and printouts.	My Company
Auto-sync interval	How often new punches are fetched from the terminal.	5 min
First sync looks back	How many days of old punches the first download collects.	30 days
Event minor codes	Which terminal events count as punches: 75 face, 1 card, 38 fingerprint. Use 0 for "all".	75,1,38
Hours before shift	Punches this long before shift start belong to that day. This keeps night shifts together.	4 h
Minimum overtime	Overtime shorter than this is ignored.	30 min
Bridge agent token	The password the bridge PC uses. Copy it into `bridge.ini`.	auto
Download backup	One file containing everything: employees, punches, settings and logins.	—

2.11 Works on your phone DONE



Every screen adapts to small screens. The menu folds into a button, and wide tables scroll sideways.

- Open the website address in Chrome or Safari and sign in.
- Tip: use **"Add to Home screen"** to get an app-like icon.
- A dedicated mobile app is FUTURE. The website covers the same needs today.

3 How attendance is calculated

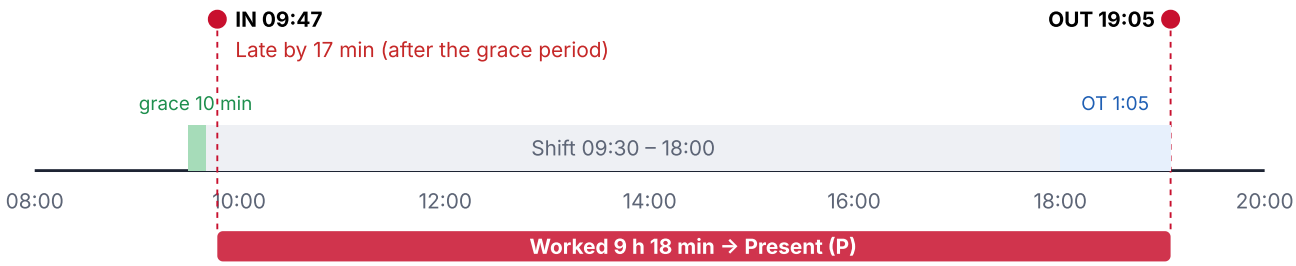
The app never stores "present" or "absent". It works them out from the raw scans and your rules each time. Change a rule and the past is recalculated too.

The seven statuses

CODE	MEANING	WHEN IT HAPPENS	PAYABLE
P	Present	In and out punches, and hours worked ≥ the shift's **full-day hours**.	1 day
HD	Half day	Hours worked are between the **half-day** and full-day minimums.	½ day
A	Absent	No punch on a working day, or fewer hours than the half-day minimum.	0
MP	Missing punch	Only one punch on a past day (forgot to punch out). Counts as present until you correct it.	1 day*
L	Leave	A leave entry covers the day and there are no punches.	1 day
H	Holiday	The date is in the Holidays list.	1 day
WO	Weekly off	The weekday is a weekly off in the employee's shift (e.g. Sunday).	1 day

* Review MP days each month and add the missing punch, so the hours and overtime are right.

A worked example: one day on the "General" shift (09:30 – 18:00)



Only the first punch (IN) and the last punch (OUT) count. Scans in between, such as a lunch break, are ignored.

Late: the first punch is after shift start + grace. The late minutes are counted from the shift start (17 min above).
Early exit: the last punch is before shift end − grace.
Overtime: time after shift end, if it is at least the minimum overtime (30 min). All hours worked on a holiday or weekly off are overtime.

Night shifts (e.g. 21:00 – 07:00): punches from 4 hours before the shift start until 24 hours later belong to that shift's day. So "in 20:46, out 07:23 next morning" is one shift of 10 h 37 m.
Payable days = Present (half day = ½) + Leave + Holidays + Weekly offs.

4 What's done

The software is complete for day-to-day attendance and ready to host. Here is everything built so far.

AREA	DELIVERED	STATUS
Terminal link	Connects to the DS-K1T320EFWX over Hikvision's official ISAPI interface with digest login. Reads punches, users and device info.	DONE VERIFY
Device actions	Send users and cards, enrol faces from a photo, delete users, sync the clock, open the door, set up real-time push.	DONE VERIFY
Bridge agent	Office-PC program for cloud hosting. Uploads punches, runs queued actions, starts automatically with Windows, logs to a file.	DONE
Attendance rules	Shifts (including night shifts), grace, breaks, half day/full day, weekly offs, holidays, leave, late, early exit, overtime.	DONE
Screens	Dashboard, daily attendance, monthly muster/summary, employee calendar, employees, shifts, holidays, leave, punch log, devices, settings, login. Employee import from the device or CSV, bulk changes, manual punches with an audit note. Works on phones.	DONE
Reports	Daily, muster roll, summary, daily detail, punch log and employee list as CSV (opens in Excel). Print layouts.	DONE
Security	Admin and viewer roles, encrypted passwords, CSRF protection, secure cookies on HTTPS, secret token for the bridge.	DONE
Safe updates & backups	Data kept outside the code, automatic database upgrades (migrations), one-command update script. Backups: download from Settings, automatic before upgrades, nightly script keeps 30.	DONE
Hosting kit	GoDaddy cPanel entry file, VPS one-command installer (HTTPS included), step-by-step hosting guide.	DONE

How it was checked

38
AUTOMATED TESTS PASSING

✔
UPDATE REHEARSAL: NO DATA LOST

0
TESTS ON A REAL TERMINAL YET

- **Attendance rules:** on time, late, grace, half day, absent, missing punch, leave, holiday, weekly off, night shift, monthly totals.
- **Terminal and bridge:** tested end to end against a *simulated* DS-K1T320 that behaves like the real ISAPI interface, including paging through large event lists and unreachable-device errors.
- **Upgrades and screens:** an update rehearsal kept the changed password, employees and punches. Every page loads, CSV exports are correct, and viewers cannot change anything.

> **Honest limitation:** no real terminal was available while building. The ISAPI commands follow Hikvision's published interface, but punches, face upload and event codes must be confirmed on your own device during the pilot (Part 5, step 8).

5 What needs to be done

To go live, work through these steps in order. Each is small, and together they take about a day. The detailed commands are in `docs/DEPLOY_GODADDY.md`.

1. **Create the `main` branch on GitHub** YOUR ACTION
   The code is on branch `ccr-9a6085eb-e2f36j`. On GitHub: Branches → New branch → name it `main`, base it on that branch, then make it the default branch.

2. **Pick the GoDaddy product** YOUR ACTION
   Use **cPanel Web Hosting** if cPanel shows **"Setup Python App"**, otherwise a **VPS**. Website Builder, Managed WordPress and Windows hosting will not work.

3. **Prepare the terminal** YOUR ACTION
   Give it a fixed IP address (router DHCP reservation). In its web page, enable **ISAPI**. Note its admin password and check its time zone.

4. **Put the app online** YOUR ACTION
   Create the sub-domain (e.g. attendance.yourcompany.com), download the code, create the Python app, install packages, turn on HTTPS. Guide Part A (cPanel) or Part B (VPS).

5. **First login and basic setup** YOUR ACTION
   Change the admin password. Set the company name, departments, shifts and this year's holidays. Create viewer logins for managers.

6. **Set up the bridge PC** YOUR ACTION
   Install Python and copy the code. Fill `bridge.ini` with the address and token from Settings, run `run_bridge.bat` once, then `install_windows_task.bat` as administrator. Set the PC to **never sleep**.

7. **Connect the terminal** YOUR ACTION
   Devices → Add device (connection: bridge) → Test connection → Import users from device → Download new punches. Assign shifts to employees.

8. **Pilot for 1–2 weeks and verify on the real device** VERIFY ON DEVICE
   Run alongside your current method and compare. Check each item in the table below.

9. **Switch on nightly backups and go live** YOUR ACTION
   Add the cron job (Guide A9), download a backup to Google Drive monthly, and stop the old method.

Pilot checklist: confirm on your DS-K1T320EFWX

✓	CHECK	IF IT DOESN'T WORK
☐	"Test connection" shows model and serial number	Check IP, password, and that ISAPI is enabled
☐	A face scan appears in the Punch log within the sync interval	Settings → Event minor codes = `0`, then Download again
☐	Punch times match the terminal's clock	Fix the device UTC offset, then "Sync device clock"
☐	"Send employee to device" creates the user on the terminal	Read the error in the job list and report it
☐	"Upload face" enrols a face that the terminal recognises	Enrol on the terminal itself meanwhile, and report the error
☐	Card swipes are recorded (if you use cards)	Make sure `1` is in the event codes
☐	Daily and monthly totals match your manual records	Adjust the shift's grace and half/full-day hours

Optional future features NOT BUILT YET

Not needed to go live. Each can be added later through the safe update process (Part 6) without losing data. The priority is a suggestion.

FEATURE	WHAT IT WOULD ADD	PRIORITY
Leave balances & approval	Yearly quotas per leave type, balance tracking, apply → approve workflow.	High
Attendance regularisation	Employees request corrections for missed punches. Managers approve, and a manual punch is added automatically.	High
Alerts	Daily email or WhatsApp/SMS summary to HR. Late or absent notifications.	High
Excel / PDF & payroll export	Formatted .xlsx/PDF muster roll and salary sheets with letterhead, and exports your payroll or Tally setup can import directly.	Medium
Employee self-service login	Employees see their own attendance and leave on their phone.	Medium
Shift roster / rotation	Different shifts on different days or weeks per employee.	Medium
Multiple sites	Several offices, each with its own terminals and bridge, plus location-wise reports.	As needed
Audit log	Who changed what and when, for every edit, not just manual punches.	Medium
Extras	Two-factor login for admins, a Hindi interface, and an installable Android/iOS app (the website already works on phones).	Low

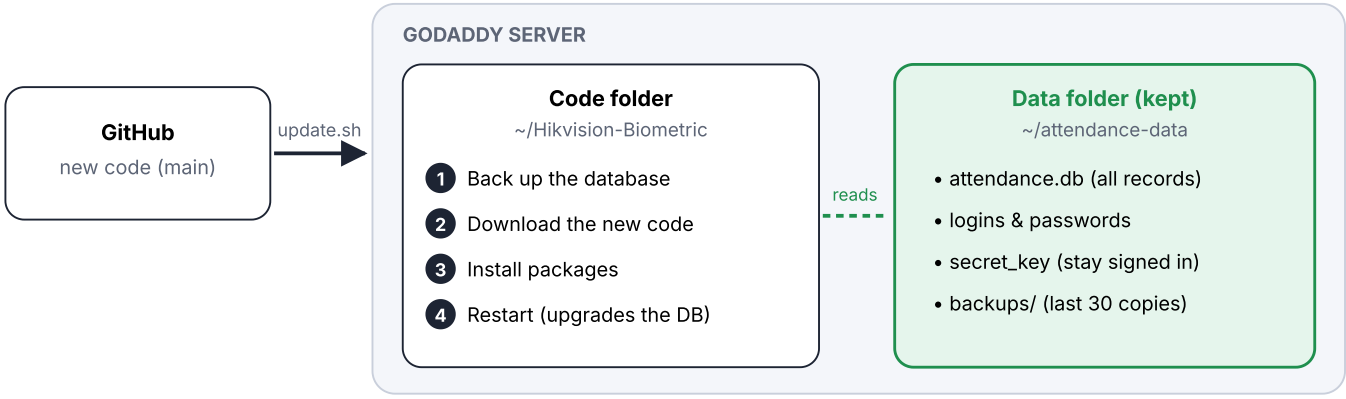
Known limitations today

- **The bridge PC must stay on.** If it is off, punches wait safely on the terminal and are uploaded when it is back. The terminal stores tens of thousands of events.
- **One shift per employee at a time.** Changing an employee's shift also changes how past days are calculated. Rotating rosters are a future feature.
- **First and last punch only.** Lunch-break in/out punches are not deducted. Use the shift's "unpaid break" setting instead.
- **Leave has no balances yet.** Entries are recorded, but quotas are not tracked.
- **Built for small and medium teams** (up to a few hundred employees) on a single database file. That is plenty for one or a few offices.

6 Keeping it running

Updates never touch your data

When a bug is fixed or a feature is added, the new code is pushed to GitHub and the server runs **one command**. The code folder is replaced, but the **data folder is never touched**.



Why nothing is lost

- The data lives in a separate folder that git never touches.
- The default admin login is only created when there are **no users at all**.
- New features upgrade the database **in place** and keep every row.
- Two backups are taken before any upgrade.

The update command

cPanel terminal:

```
bash ~/Hikvision-Biometric/deploy/update.sh
```

VPS:

```
sudo bash /opt/Hikvision-Biometric/deploy/update.sh --systemd
```

Only update the bridge PC if the release notes say so.

Routine care

Daily (2 min)

- Dashboard: is the device dot green?
- Check "Late today" and absentees.

Weekly (10 min)

- Fix MP (missing punch) days.
- Enter the week's leave.
- Devices: any failed jobs?

Monthly (15 min)

- Download the Summary CSV for payroll.
- Settings → Download backup and store it off-site.
- Add next month's holidays.

Restoring a backup

In cPanel → Setup Python App, click **Stop**. Copy the backup file over `~/attendance-data/attendance.db` (keep the old one as a spare), then click **Start**. The exact commands are in the hosting guide, section 4.

7 Help & glossary

Quick fixes

PROBLEM	WHAT TO DO
Device shows "last check-in: never"	The bridge isn't running. On the office PC, double-click `bridge\run_bridge.bat` and read the message. Check the PC isn't asleep.
Jobs fail: "Cannot reach device"	On the bridge PC, open the terminal's IP in a browser. If it doesn't load, the IP changed or the PC is on another network.
Jobs fail: "Authentication failed"	The terminal password is wrong. Fix it in Devices → Edit.
Bridge: "Server rejected the bridge token"	Copy the token from Settings → Bridge agent into `bridge.ini` again, then restart the bridge.
No punches although scans work	Settings → Event minor codes = `0`, then "Download new punches".
Times are a few hours off	Set the device's UTC offset (+05:30 for India), then click "Sync device clock".
Someone shows Absent but was present	Check the Punch log for that day. If the scan is missing, add a manual punch.
Website shows "It works!" or 503	See Troubleshooting in `docs/DEPLOY_GODADDY.md` (usually a missed install step).
Forgot the admin password	In the server terminal: `flask --app app reset-admin` (with the app's environment active). The password becomes `admin`; change it immediately.

Glossary

Punch	One scan (face or card) at the terminal, with date and time.
Terminal / device	The Hikvision DS-K1T320EFWX on the wall.
ISAPI	Hikvision's official way for software to talk to its devices. It must be enabled on the terminal.
Bridge / bridge agent	The small program on an office PC that connects the terminal to the online app.
Job	An action you clicked (e.g. "open door") waiting for the bridge to carry it out.
Grace period	Minutes allowed after shift start before someone is marked late.
Muster roll	Month grid of every employee's status for every day.
Payable days	Days that count for salary: present + leave + holidays + weekly offs (half day = ½).
CSV	A simple spreadsheet file that opens directly in Excel or Google Sheets.
Migration	An automatic, data-preserving upgrade of the database that ships with a new version.
cPanel / VPS	Two kinds of GoDaddy hosting: a managed control panel, or your own virtual server.

Where everything lives in the code repository

`docs/DEPLOY_GODADDY.md` hosting guide · `README.md` technical overview · `bridge/` office-PC program · `deploy/` update, backup and VPS scripts · `tests/` automated checks.